

Ahmet Kurt, Ph.D.

Assistant Professor

Bryan, TX, USA

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🔗 [Google Scholar](#)

🌐 [LinkedIn](#)

🐙 [GitHub](#)

Summary

- Interests Cybersecurity, Blockchain, Wireless Networks, IoT Security, Distributed Systems.
- Programming Python, C++, C, Bash, Rust, JavaScript, Go.
- Skills Linux, LaTeX, Wireshark, Core Lightning, Ind, Rust Lightning, Bitcoin Core, PyCharm, CLion, VSCode, AWS, GCP, Matplotlib, NumPy, SciPy, TensorFlow, Keras, NetworkX, Pandas.

Education

- 2019–2023 **Ph.D. in Electrical and Computer Engineering**, *Florida International University*, Miami, FL, GPA: 3.92/4.00
Dissertation: *Enabling Third Layer Bitcoin Applications Using Lightning Network* [🔗](#)
Advisor: Prof. Kemal Akkaya
- 2019–2021 **M.Sc. in Electrical and Computer Engineering (en route to Ph.D.)**, *Florida International University*, Miami, FL
- 2013–2018 **B.Sc. in Computer Engineering**, *Antalya Bilim University*, Antalya, Turkey
B.Sc. in Electrical and Electronics Engineering, *Antalya Bilim University*, Antalya, Turkey

Media Appearances

- Featured at [Forbes](#) [🔗](#), [Decrypt](#) [🔗](#) [🔗](#), [FIU News](#) [🔗](#), [TFTC](#) [🔗](#), [ZenGo](#) [🔗](#) and many other smaller sites.

Patents

- US Patent Systems and Methods for Detecting Cryptocurrency Wallet Artifacts in a File System [🔗](#)

Work Experience

- Aug 2024–Present **Assistant Professor**, *East Texas A&M University*, Bryan, TX
Joined the Computer Science and Information Systems department (RELLIS campus).
 - Teaching courses, conducting research and performing service.
 - Courses Taught:** CSCI 451 (Wireless and Mobile Security), CSCI 463 (Systems Security & Trusted Computing), CSCI 399 (Junior Cyber Design Project), CSCI 499 (Senior Cyber Design Project), CSCI 458 (Network Security & Management), CSCI 459 (AI Enhanced Security), CSCI 353 (Threat and Vulnerability Management).
- Jun–Sep 2024 **Security Apprenticeship Program [PhD]**, *Nethermind*, Remote
 - Worked on Ethereum smart contract security audits within the Nethermind security team.
- 2023–2024 **Postdoctoral Associate**, *Florida International University*, Miami, FL
 - Conducted research on blockchain, security, and wireless network projects.
- 2019–2023 **Graduate Research and Teaching Assistant**, *Florida International University*, Miami, FL
Key Projects:
 - LNMesh:** Implemented true offline Bitcoin Lightning payments using WiFi & Bluetooth; created a Python simulator. Publication: [🔗](#)
 - LNGate & LNGate²:** Enabled secure LN payments for resource-constrained devices using threshold cryptography. Publications: [🔗](#), [🔗](#)
 - LNBot & D-LNBot:** Designed a covert hybrid botnet on the Lightning Network to demonstrate security vulnerabilities. Publications: [🔗](#), [🔗](#)
 - Crypto Wallet Forensics:** Designed and implemented a Machine Learning-based artifact detection framework for crypto wallets. Publications: [🔗](#), [🔗](#)
 - Ns-3 Work:** Modified ns-3 IEEE 802.11s HWMP protocol in C++ for distributed connectivity maintenance. Publications: [🔗](#), [🔗](#)

Teaching: Instructor for EEL 3712L (Logic Design I Lab); Main TA for EEL 3110 (Circuit Analysis).

2016–2017 **Embedded Software Engineer**, *Desird Design R&D*, Antalya, Turkey
 - Developed software for a product using ESP8266 WiFi Modules in C++ and Lua.

- 2017 **Intern, Havelsan**, Istanbul, Turkey
 - Developed a desktop GUI application using Qt in C++.
- 2016 **Erasmus+ Intern, University of Bristol**, Bristol, UK
 - Contributed to 'optically reconfigurable microwave switches' research project.

Professional Service

- 2026 Publicity Chair, IEEE IPCCC 2026 (45th IEEE International Performance Computing and Communications Conference), Austin, TX. [🔗](#)
- 2026 Panelist, National Science Foundation (NSF), CyberAI Corps Scholarship for Service (CyberAI SFS) Program.
- 2026 Panelist, National Science Foundation (NSF), Secure and Trustworthy Cyberspace (SaTC) Program.
- 2026 TPC Member, IEEE LCN 2026 (51st IEEE Conference on Local Computer Networks), Coimbra, Portugal.
- 2026 TPC Member, SAFE-ML 2026 Workshop (co-located with IEEE ICST 2026), Daejeon, South Korea. [🔗](#)
- 2025 TPC Member, Emerging Technologies toward Intelligent Systems Workshop (co-located with IEEE IPCCC 2025), Austin, TX. [🔗](#)

Honors and Awards

- 2026 AI in Cybersecurity Education - Faculty Development Summer Institute 2026 [🔗](#)
- 2023 FIU Dissertation Year Fellowship [🔗](#)
- 2021 Black Hat USA 2021, Student Scholarship [🔗](#)
- 2020 RSA 2020 Conference, Security Scholar [🔗](#)

Selected Publications

- 2026 T. Bakaysa, **A. Kurt**, A. Beibitkhan, J. M. Romo Diaz de Leon, T. Kalat, J. Kramer, E. Rodriguez, A. Watkins, A. Aydeger. *LNTest: A Testbed for Evaluating Bitcoin Lightning Network-Based Botnets*, in **21st International Conference on Availability, Reliability and Security (ARES 2026)** (accepted).
- 2026 A. Aydeger, E. Zeydan, **A. Kurt**. *AI-Enabled Cloud-Edge-IoT Continuum for Hyper-Distributed and Trustworthy Applications*, in **IEEE 35th Wireless and Optical Communications Conference (WOCC 2026) - Artificial Intelligence and Big Data Analytics Symposium** (to appear).
- 2025 A. Bhattarai, A. Sahin, M. Veksler, **A. Kurt**, D. Aras, C. Imery, K. Akkaya. *Cryptocurrency Forensics Automation: A Deep Learning and NLP-based Approach for Mobile Platforms*, in **Discover Computing**. [🔗](#)
- 2024 **A. Kurt**, E. Erdin, K. Akkaya, A. S. Uluagac, M. Cebe. *D-LNBot: A Scalable, Cost-Free and Covert Hybrid Botnet on Bitcoin's Lightning Network*, in **IEEE Transactions on Dependable and Secure Computing**. [🔗](#)
- 2024 **A. Kurt**, K. Akkaya, S. Yilmaz, S. Mercan, O. Shlomovits, E. Erdin. *LNGate²: Secure Bidirectional IoT Micro-payments Using Bitcoin's Lightning Network and Threshold Cryptography*, in **IEEE Transactions on Mobile Computing**. [🔗](#)
- 2023 **A. Kurt**, A. Sahin, R. Harrilal-Parchment, K. Akkaya. *LNMesh: Who Said You Need Internet to Send Bitcoin? Offline Lightning Network Payments Using Community Wireless Mesh Networks*, in **IEEE International Symposium on a World of Wireless, Mobile and Multimedia Networks (WoWMoM 2023)**. [🔗](#)
- 2022 S. Mercan, **A. Kurt**, E. Erdin, K. Akkaya. *Cryptocurrency Solutions to Enable Micro-payments in Consumer IoT*, in **IEEE Consumer Electronics Magazine**. [🔗](#)
- 2021 **A. Kurt**, S. Mercan, O. Shlomovits, E. Erdin, K. Akkaya. *LNGate: Powering IoT with Next Generation Lightning Micro-payments Using Threshold Cryptography*, in **ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec '21)**. [🔗](#)
- 2021 **A. Kurt**, S. Mercan, E. Erdin, K. Akkaya. *Enabling Micro-payments on IoT Devices Using Bitcoin Lightning Network*, in **IEEE International Conference on Blockchain and Cryptocurrency (ICBC 2021)**. [🔗](#)
- 2021 **A. Kurt**, N. Saputro, K. Akkaya, A. S. Uluagac. *Distributed Connectivity Maintenance in Swarm of Drones During Post-Disaster Transportation Applications*, in **IEEE Transactions on Intelligent Transportation Systems**. [🔗](#)
- 2020 **A. Kurt**, E. Erdin, M. Cebe, K. Akkaya, A. S. Uluagac. *LNBot: A Covert Hybrid Botnet on Bitcoin Lightning Network for Fun and Profit*, in **European Symposium on Research in Computer Security (ESORICS 2020)**. [🔗](#)

Languages

Turkish Native
English Proficient